

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru-3bb728174 | github.com/zac-09 | Kampala, Uganda (UTC+3) — Remote

PROFESSIONAL SUMMARY

Senior Backend Engineer with 6+ years building high-throughput, correctness-critical systems in Node.js and JavaScript/TypeScript. Led the migration of 20+ production applications from Parse/MongoDB to Firebase/GCP with zero downtime while the system stayed live throughout — and built the pipeline that made it possible: an idempotent, real-time two-way MongoDB↔Firestore sync on Google Pub/Sub that held two datastores consistent under parallel production traffic on an at-least-once bus, with no data loss and a rollback option open for the entire cutover. That work is the same problem shape as money movement: redelivery-safe processing, retries that never duplicate an effect, and reconciliation you can audit. Also builds event-driven backends on Apache Kafka, NATS Streaming and gRPC, and cuts cost structurally — \$5,000/month eliminated by consolidating an estate off AWS. Proficient in Node.js, JavaScript/TypeScript, SQL and MongoDB; intermediate Go and Rust. Remote-first since 2020 across US, Singapore and African teams.

CORE COMPETENCIES

Distributed Systems at Scale

Idempotency & Exactly-Once Processing

Event Pipelines (Kafka · Pub/Sub · NATS)

Data Consistency & Reconciliation

SQL & NoSQL Data Modelling

Node.js Services Under Load

Zero-Downtime Migrations

Cost & Margin Optimisation

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer — US Remote

- **Led the end-to-end migration of 20+ production applications** from Parse/MongoDB to Firebase/GCP with zero downtime, reporting directly to the CTO — service-by-service cutover with a rollback path held open throughout.
- Built an **idempotent, real-time two-way sync between MongoDB and Firestore** using Node.js and Google Pub/Sub — redelivery-safe consumers on an at-least-once bus with conflict handling, keeping both datastores consistent while each served live production traffic. Zero data loss.
- Consolidated the infrastructure estate off AWS (S3/EC2/EBS) onto GCP, **eliminating \$5,000/month** in recurring cost; migrated file storage from S3 to GCS via a Python script, sequencing cutovers so nothing user-facing broke.
- Implemented Express Shipping end to end on a revenue-critical path, generating an additional **\$40 per order**.
- Built a 3D visualisation feature using video overlay and the ffmpeg library, increasing buyer conversion.

Stack: Node.js, JavaScript, Firebase/Firestore, GCP Pub/Sub, MongoDB, AWS S3/EC2/EBS, Python, ffmpeg, React

MTailor

Mar 2022 – Jul 2022

Contractor (Backend Migration) — US Remote

- Hired onto an unfamiliar stack (Firebase/GCP) and ramped while shipping: built the MongoDB→Firestore migration scripts, running complex transformation logic over production data under contract scrutiny.
- Wrote the **Firebase SDK documentation** (Node.js and Python) that fellow engineers onboarded from, and trained the Ops team on the new dashboard and backup procedures — converted to full-time and handed leadership of the whole migration.

Stack: Node.js, Python, Firebase, GCP, MongoDB, AWS, JavaScript

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer — Singapore Remote

- Led optimisation of a real-time stock-price PWA: built batched jobs querying **2M+ Firestore records** against Morningstar APIs, reshaping query patterns and read paths rather than scaling hardware to keep customer-facing prices current under cost and latency pressure.
- Extended the backend API surface with Firebase Cloud Functions and Express.js; added responsive React/Tailwind pages to the PWA.

Stack: Node.js, Firebase, Express.js, React, JavaScript

Mind2matter

Jan 2022 – Jul 2022

Software Engineer — US Remote

- Built backends for **DeFi applications** using Web3 and Node.js, where value movement on-chain is irreversible — validate before signing, make retries safe, keep the API contract stable for the front end. Reported directly to the CTO.
- Delivered React UIs from Figma designs under tight agency deadlines alongside the backend work.

Stack: Node.js, Web3, Firebase, React, JavaScript

CodeBits

Jan 2020 – Jul 2021

Software Engineer (Team Lead) — Uganda Remote

- **Architected an event-driven microservices backend** for FIDA Uganda's case management system using Apache Kafka, Docker and Kubernetes — designed the service and event boundaries, and led a team of 4 developers through delivery.
- Built a USSD service for legal-aid providers talking to a Node.js backend over **gRPC**, with a versioned, provider-agnostic protobuf contract that multiple third parties integrated against with no bespoke per-provider code.
- Delivered a case management system for FIDA Uganda and a paralegal database for LASPNET, plus a cross-platform React Native mobile app with Firebase push notifications.

Stack: Node.js, Apache Kafka, NATS Streaming, gRPC, Docker, Kubernetes, Firebase, React Native, JavaScript

PROJECTS

Real-time Two-way Sync Pipeline (MTailor)

Message-driven MongoDB↔Firestore synchronisation backbone that supported parallel production traffic on two datastores for months during a phased migration. Built idempotent from day one because Pub/Sub delivers at least once — dedup keys, conflict handling, and safe retries throughout. Zero data loss.

Node.js · Google Pub/Sub · MongoDB · Firestore

Kafka/Kubernetes Event-Driven Platform (FIDA Uganda)

Production event-driven microservices backend serving web, mobile and USSD clients — event boundaries designed to avoid the distributed-monolith trap, deployed on Kubernetes with Docker and a versioned gRPC contract for external integrators.

Apache Kafka · Docker · Kubernetes · gRPC · NATS Streaming · Node.js

career-ops [Open Source](#)

Agentic job-search pipeline built on Claude Code: live Playwright verification, YAML configuration, Node.js PDF generation and batch worker agents. Open-sourced; used to evaluate 100+ roles end to end.

Node.js · Playwright · YAML · LLM agents

EDUCATION

BSc Computer Engineering — [Makerere University](#), Kampala

2018 – 2022

SKILLS

Proficient: Node.js, JavaScript/TypeScript, SQL (PostgreSQL), MongoDB, Firebase/Firestore, GCP, Python, Git, Linux

Intermediate: Go, Rust, AWS, React Native

Distributed Systems: Apache Kafka, Google Pub/Sub, NATS Streaming, gRPC, Docker, Kubernetes, idempotent message processing

Frontend: React, Tailwind, HTML, CSS **Languages:** English, Kiswahili